# Prompting Large Language Models

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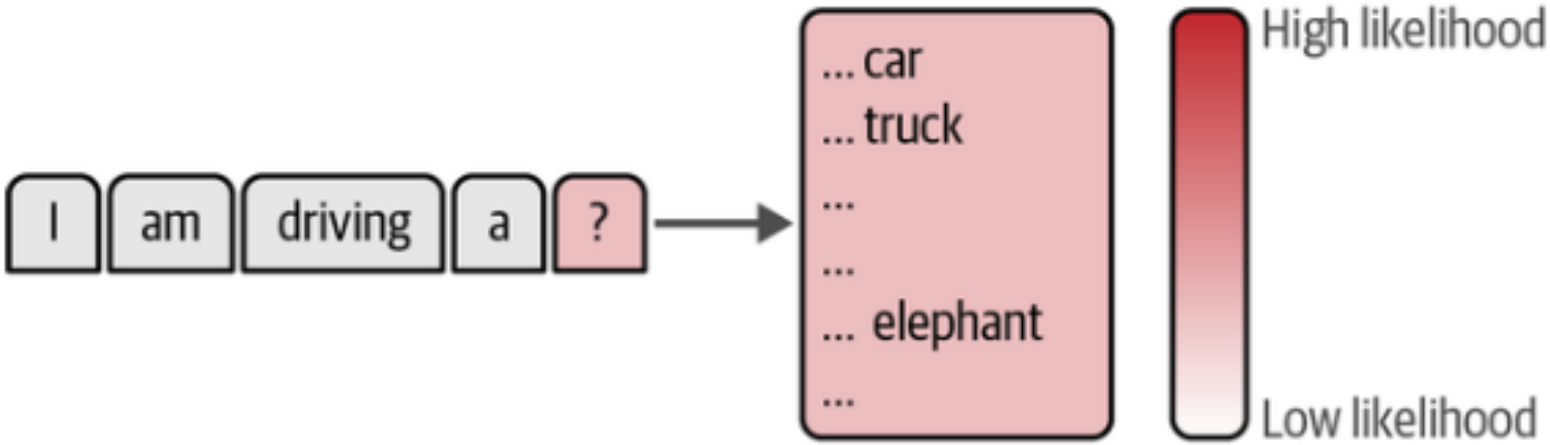
# Token Probabilities

- LLM predicts next token
- Each token has probability score
- Sampling determines creativity



# Temperature vs Top-p: What Do They Actually Do?

- Both modify token sampling from the model's probability distribution
- Temperature rescales logits before softmax
- Top-p truncates the candidate token set based on cumulative probability

**Temperature (T)**

$$P_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

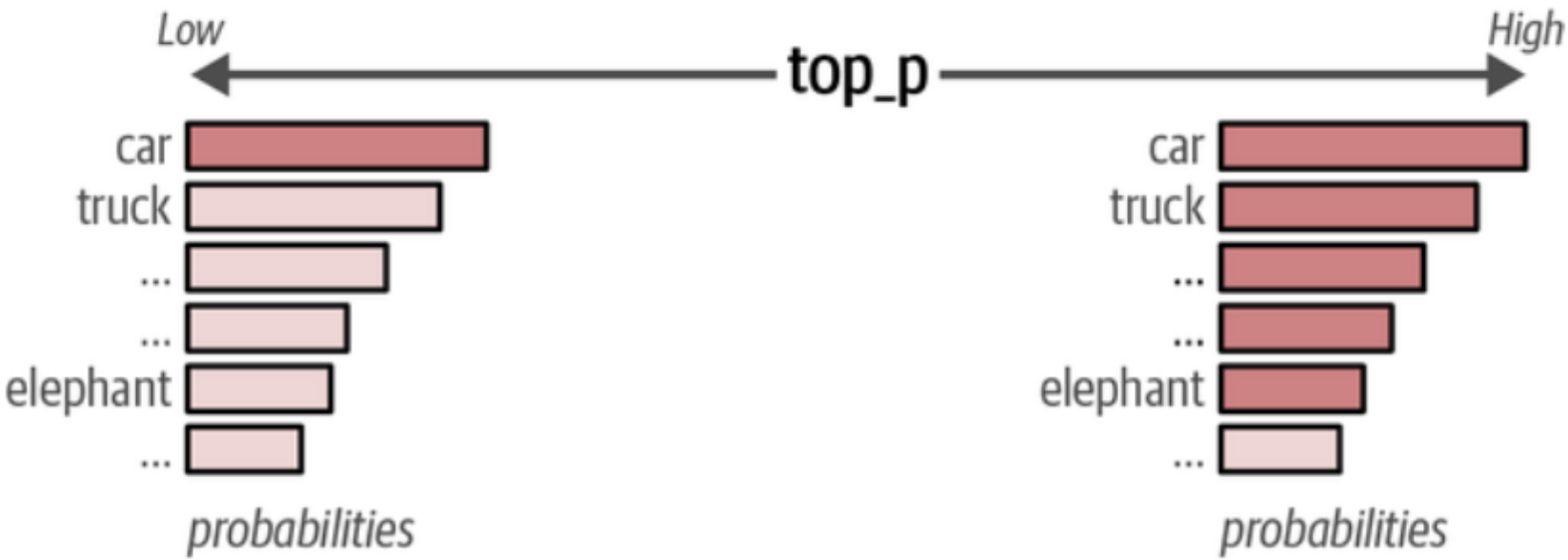
- $T < 1$: sharper distribution $\rightarrow$ more deterministic
- $T > 1$: flatter distribution $\rightarrow$ more diverse

# Temperature in LLM Generation

- Temperature $T$ scales logits before softmax: $p_i \propto \exp(_i / T)$.
- Low $T$ (0–0.5) yields deterministic outputs, focusing on high–probability tokens.
- Moderate $T$ (0.5–1.0) provides balanced diversity and coherence.
- High $T$ ($> 1$) increases randomness, potentially generating creative but less reliable responses.
- $T = 0$ corresponds to greedy decoding without sampling.
- Choose $T$ based on the desired trade–off between creativity and consistency.

# Nucleus Sampling (Top-p)

- Sort tokens by probability (descending)
- Select smallest set where cumulative probability $\geq p$
- Sample only from this nucleus



# Practical Comparison

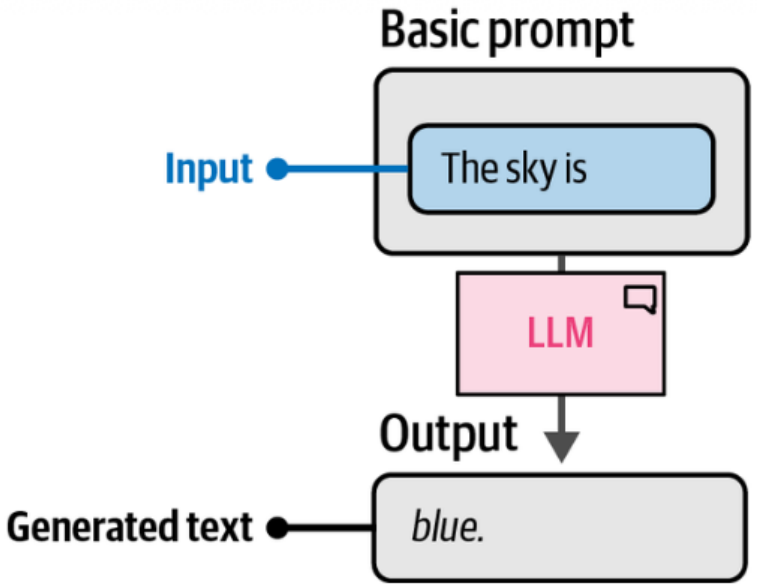
| Use Case | Temperature | Top-p |
|---|---|---|
| Deterministic QA | 0.0–0.3 | 0.1–0.5 |
| Email writing | 0.3–0.6 | 0.5–0.8 |
| Creative writing | 0.8–1.2 | 0.8–1.0 |
| Brainstorming | 1.0+ | 0.9–1.0 |

**Key insight:**

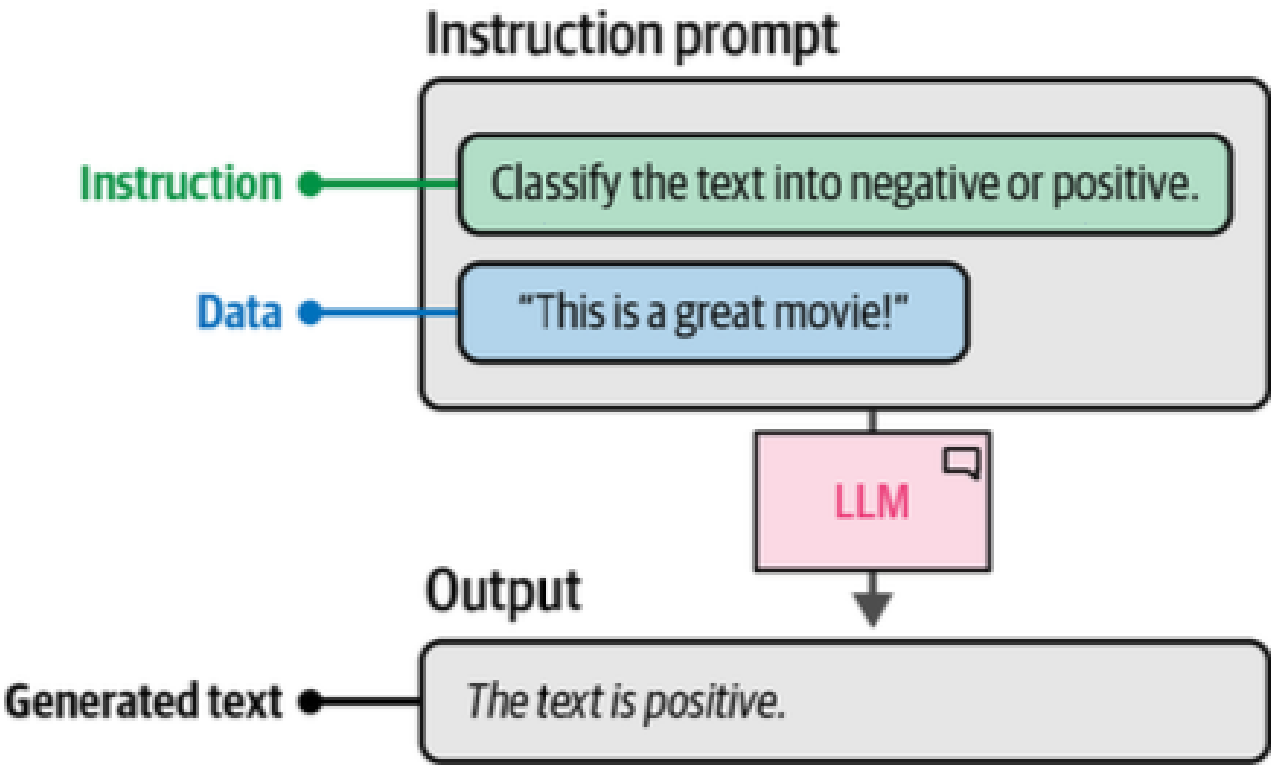
- Temperature changes *shape* of distribution
- Top-p changes *size* of candidate pool
- They interact but are not equivalent

# Ingredients of a Prompt

- Designing inputs to guide LLM output
- Iterative optimization process
- No perfect prompt exists



# Instruction Prompt

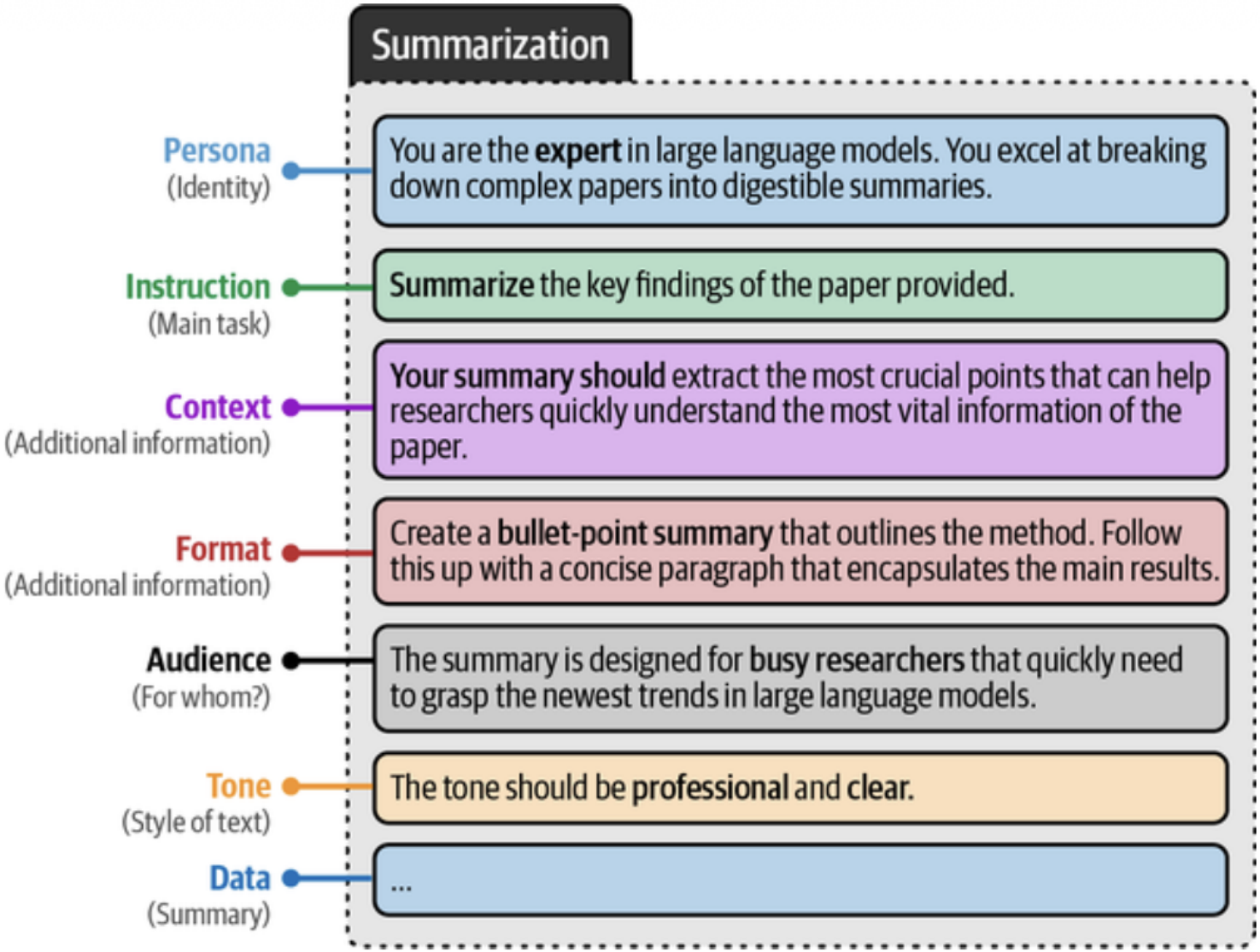


# Output Indicators

- Guide structured responses
- Example:

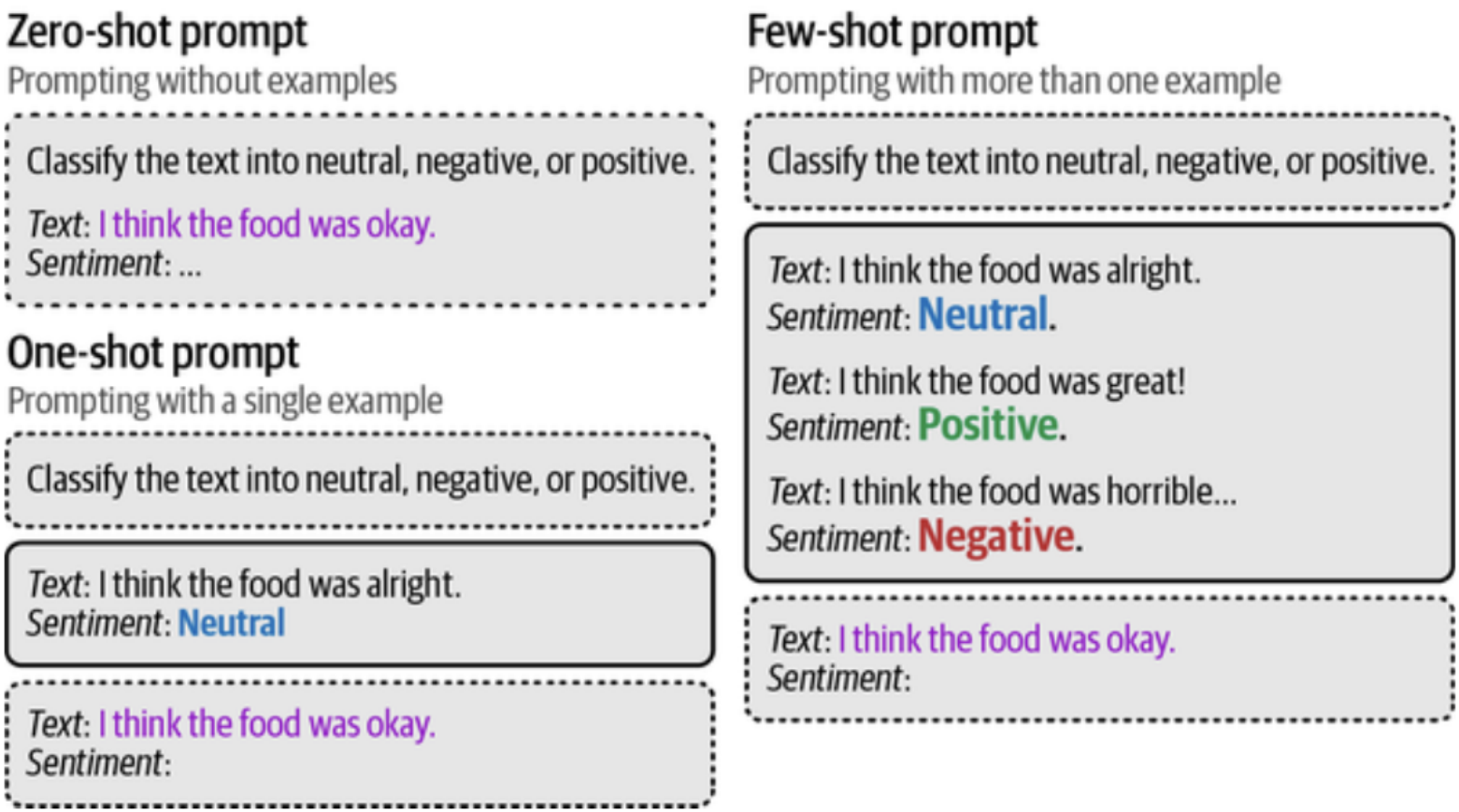
```
Text: I love this movie.
Sentiment:
```

# Modular Prompt Components

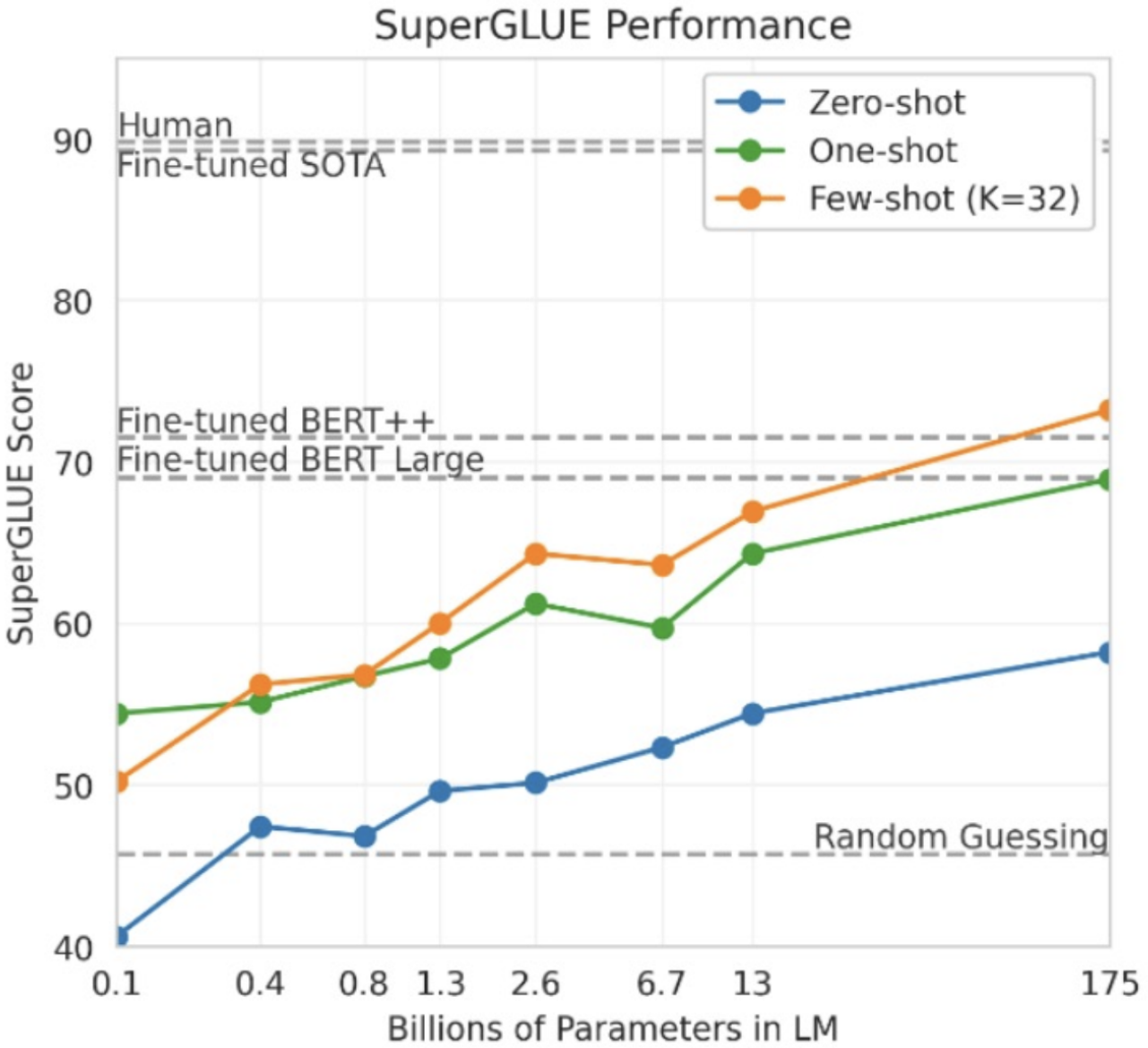


# Zero-shot, One-shot, Few-shot

- Zero-shot: no examples
- One-shot: single example
- Few-shot: multiple examples



# Comparison of fine-tuned vs prompted



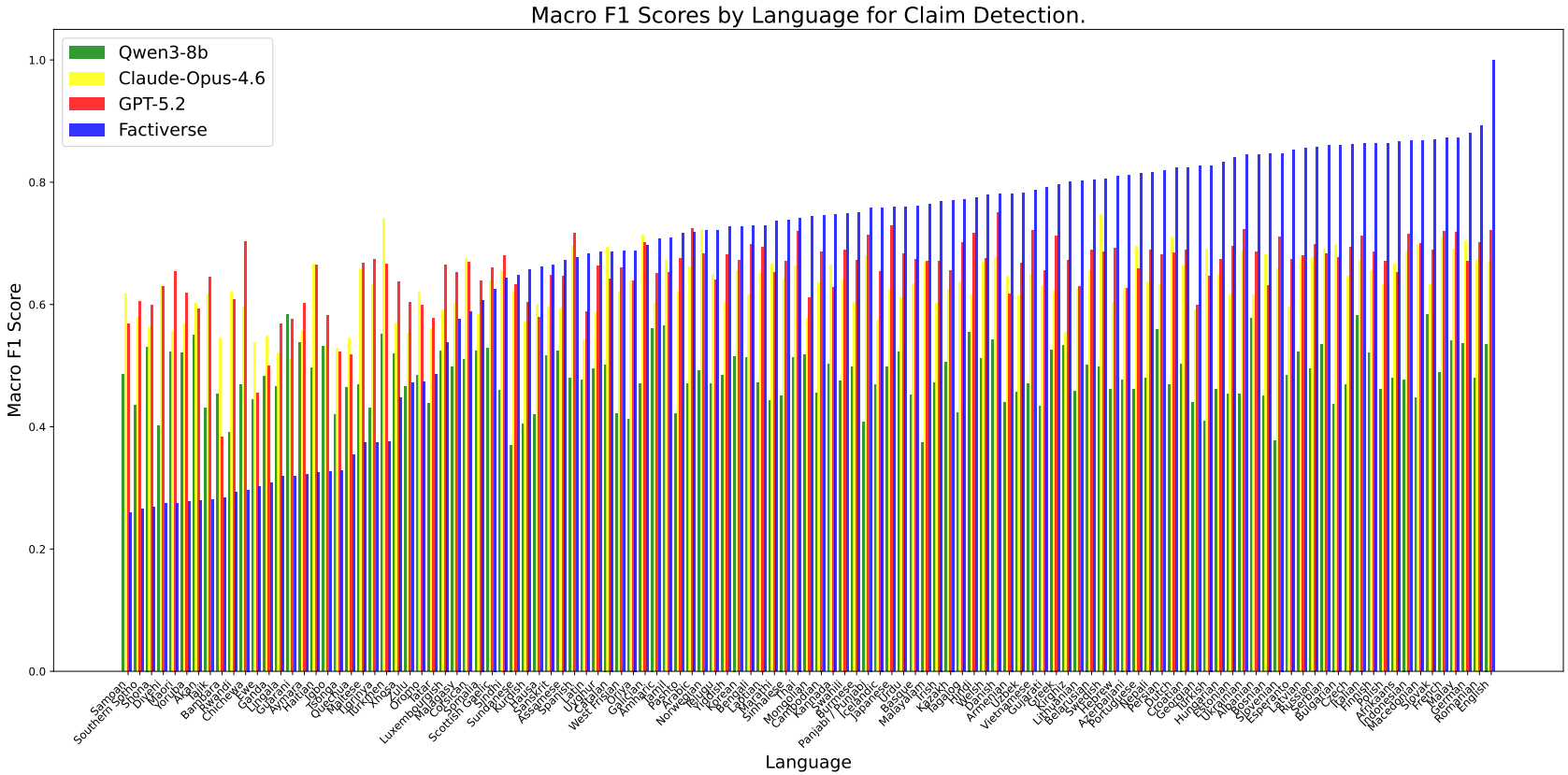

Brown et al. 2020

# Comparison of fine-tuned vs prompted (machine translation)

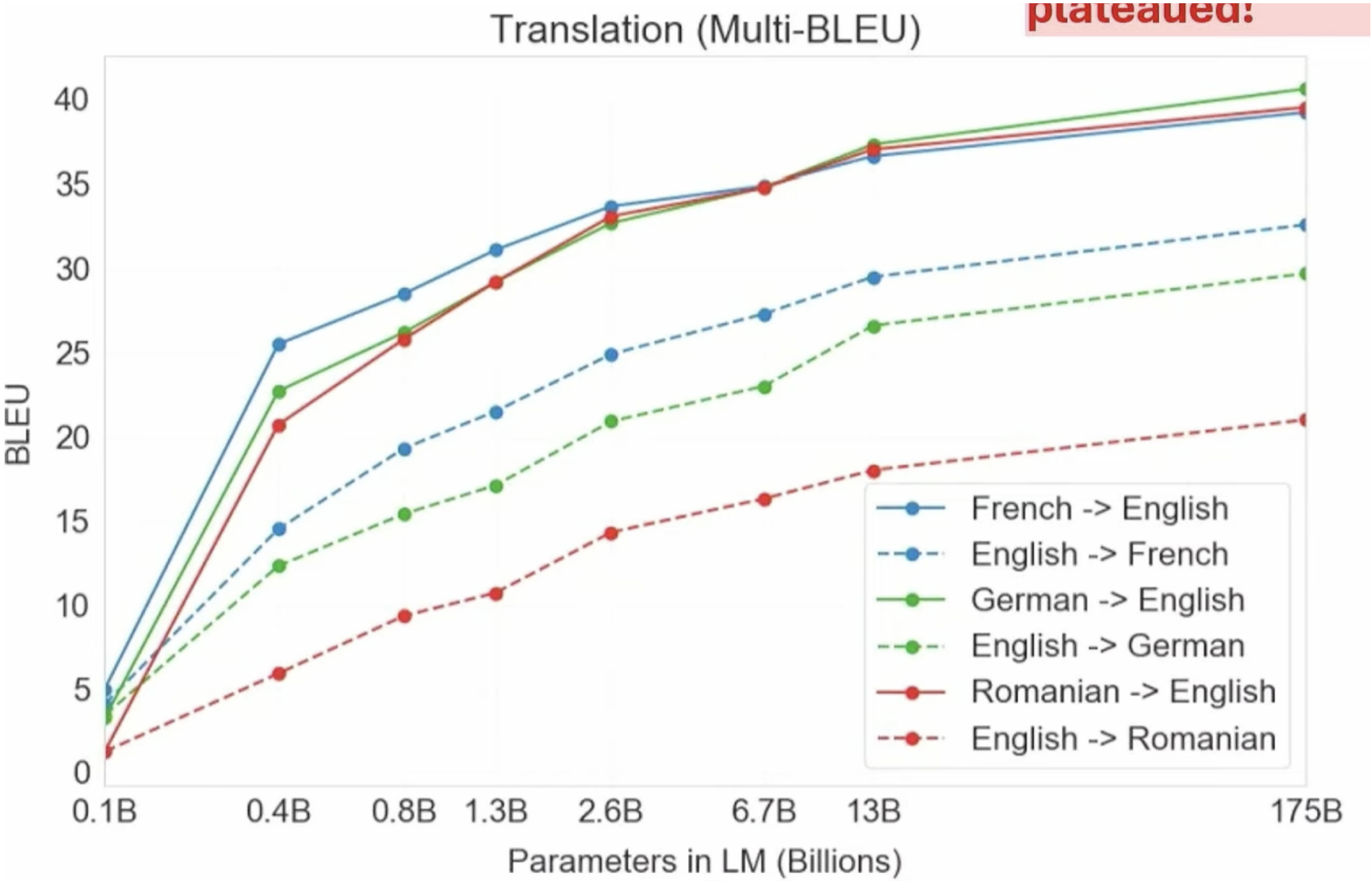
| Setting | En→Fr | Fr→En | En→De | De→En | En→Ro | Ro→En |
|---|---|---|---|---|---|---|
| SOTA (Supervised) | **45.6**[a] | 35.0 [b] | **41.2**[c] | 40.2[d] | **38.5**[e] | **39.9**[e] |
| XLM [LC19] | 33.4 | 33.3 | 26.4 | 34.3 | 33.3 | 31.8 |
| MASS [STQ+19] | 37.5 | 34.9 | 28.3 | 35.2 | 35.2 | 33.1 |
| mBART [LGG+20] | - | - | 29.8 | 34.0 | 35.0 | 30.5 |
| GPT-3 Zero-Shot | 25.2 | 21.2 | 24.6 | 27.2 | 14.1 | 19.9 |
| GPT-3 One-Shot | 28.3 | 33.7 | 26.2 | 30.4 | 20.6 | 38.6 |
| GPT-3 Few-Shot | 32.6 | 39.2 | 29.7 | 40.6 | 21.0 | 39.5 |

Brown et al. 2020

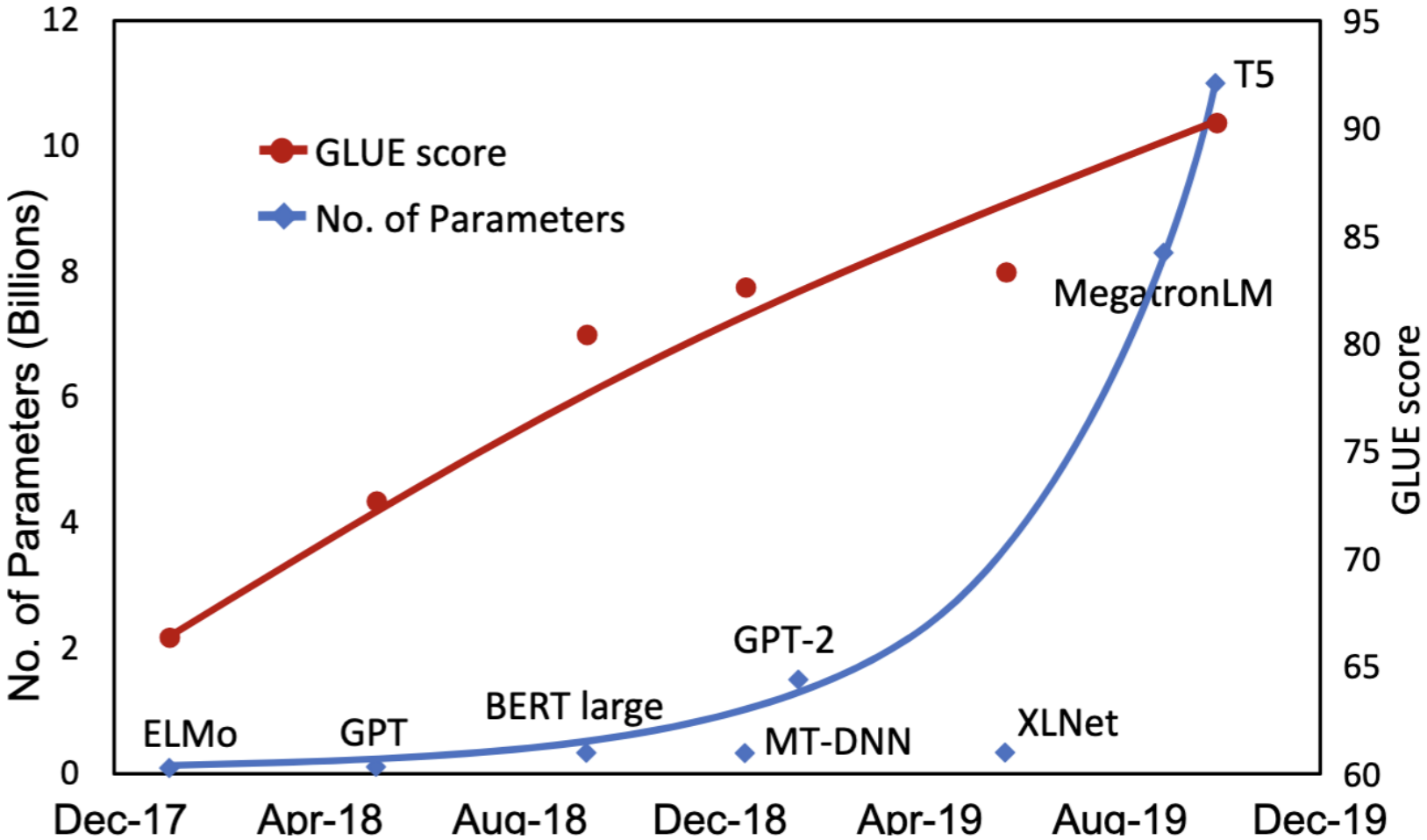
# Comparison of fine-tuned vs prompted (claim detection task)



# Model size scaling



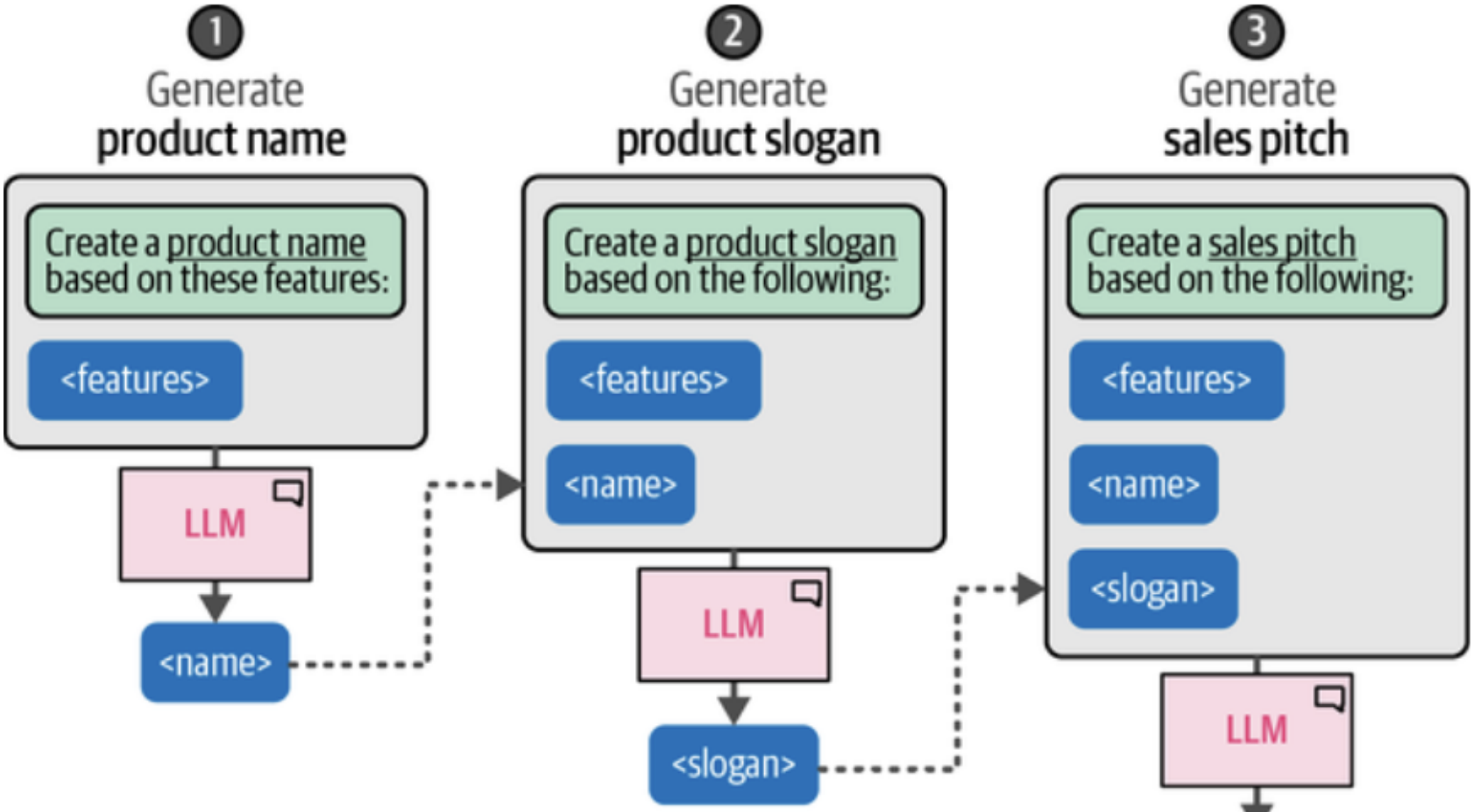
# Model size limitation




Ahmet et al. 2020

# Chain Prompting

- Break complex tasks into steps
- Output of step i becomes input of step i+1
- Improves modularity and control

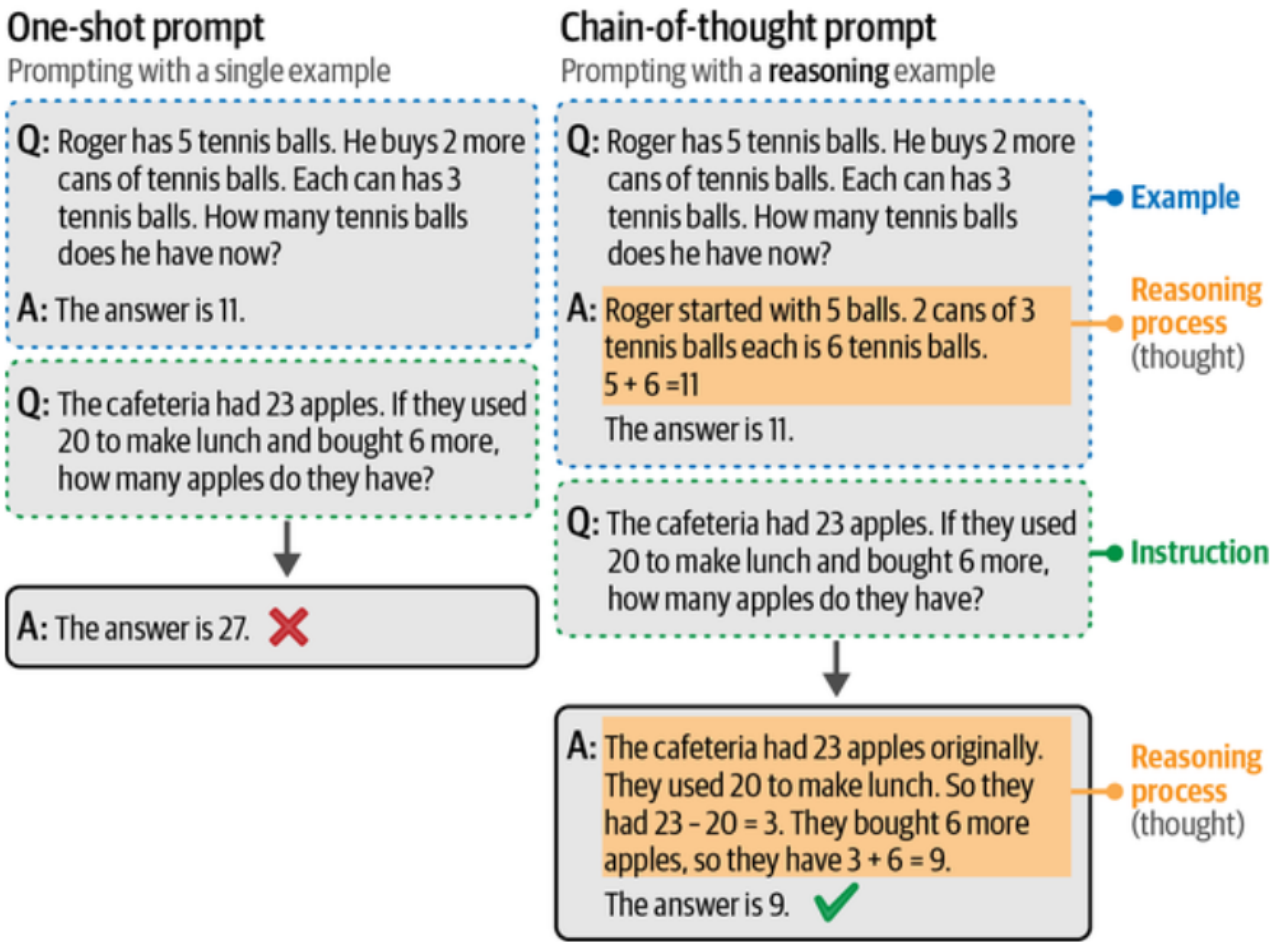


# System 1 vs System 2

- "Think fast and slow" Kahneman 2011
- "System 1": fast, intuitive (thinking represents an automatic, intuitive, and near-instantaneous process. )
- "System 2": slow, deliberate (thinking is a conscious, slow, and logical process, akin to brainstorming and self-reflection.)
- Reasoning models mimic "system 2"

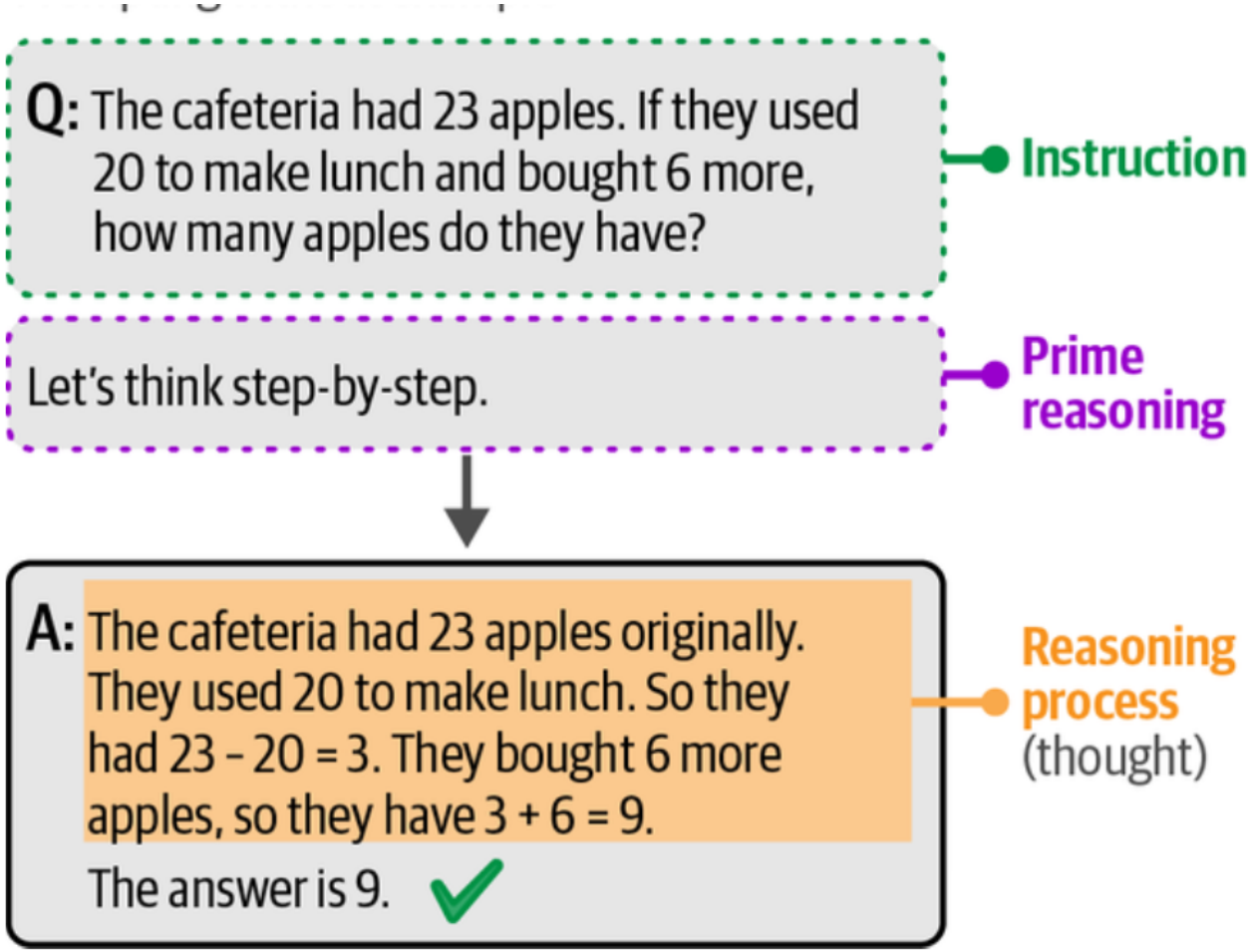
# Chain-of-Thought

- Encourage step-by-step reasoning
- Improves multi-step tasks



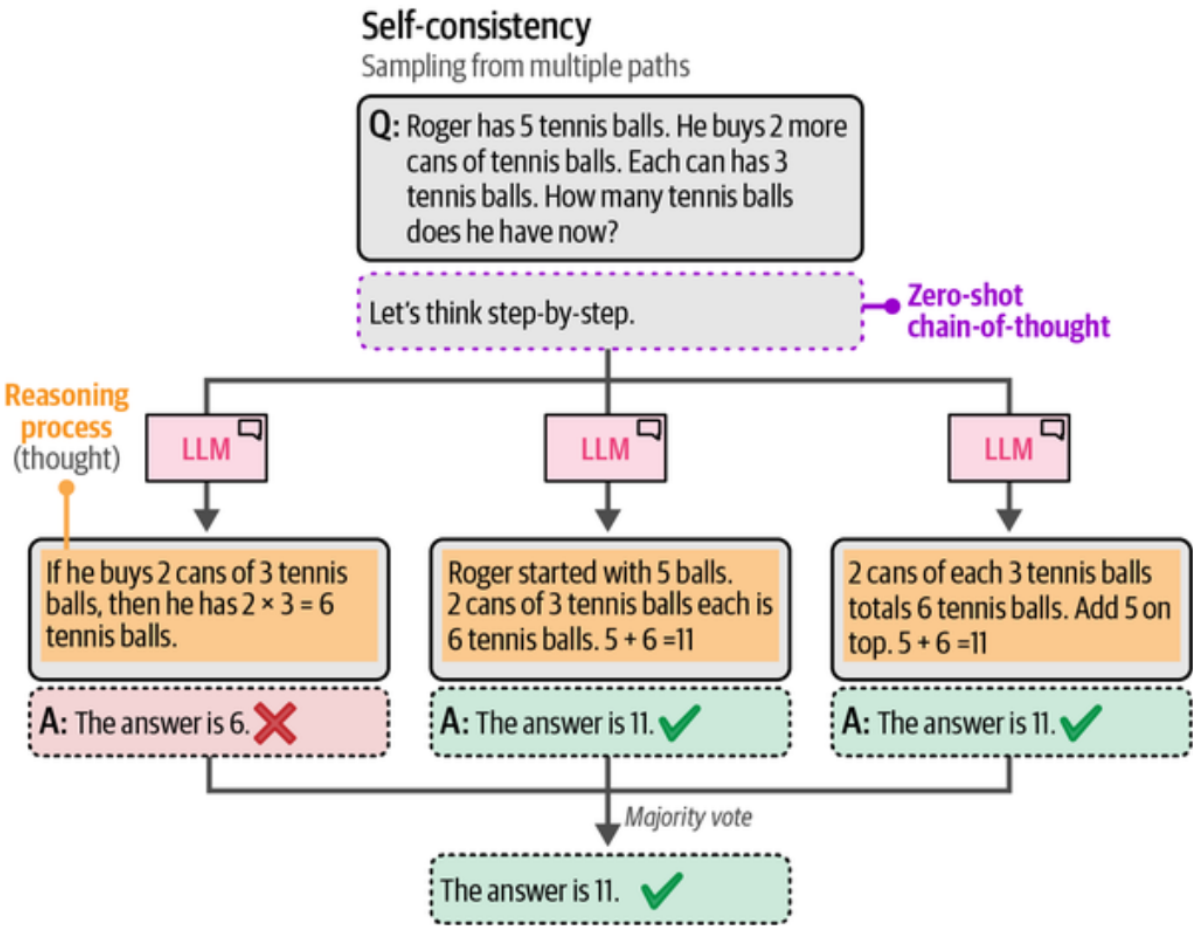
# Zero-shot Chain-of-Thought

- Add phrase: "Let's think step-by-step"
- No examples required



# Self-Consistency

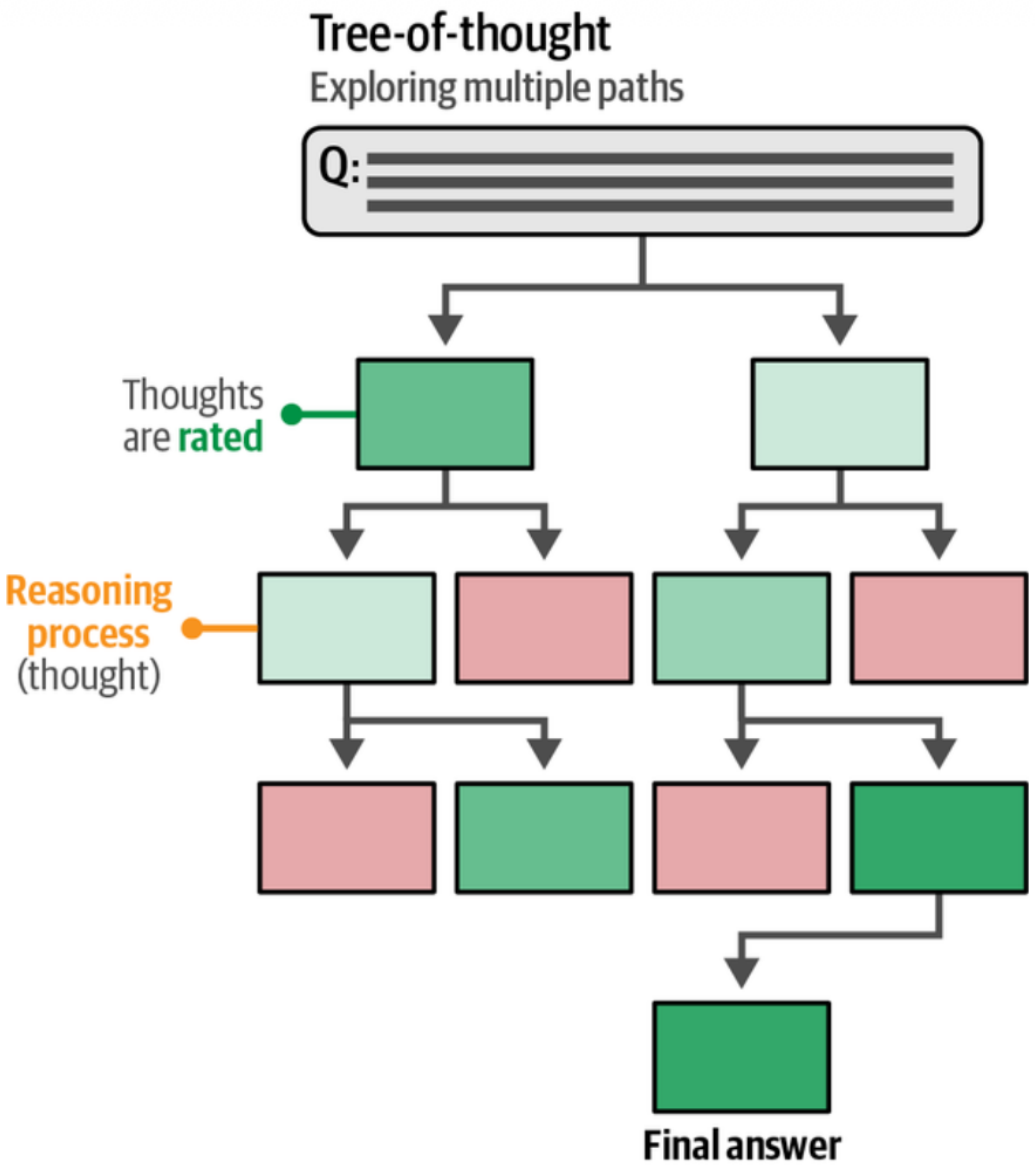
- Sample multiple reasoning paths
- Majority voting on final answer
- Improves robustness



# Tree-of-Thought

- Explore multiple intermediate thoughts
- Score and prune branches
- Useful for planning and creativity

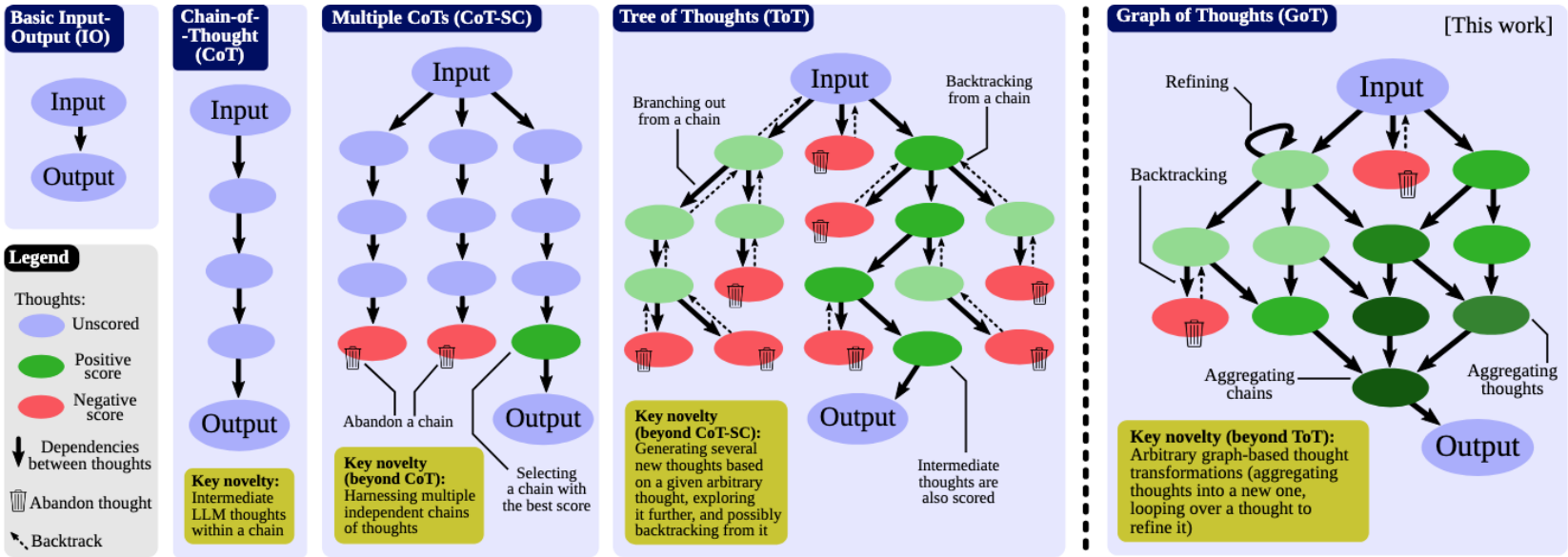
# Tree-of-Thought (cont.)



# Graph-of-Thought

- Explore multiple intermediate thoughts
- Score and prune branches
- Useful for planning and creativity

# Graph-of-Thought (cont.)




Besta et al. 2024

# Why Verify Output?

- Structured output (JSON)
- Valid labels
- Ethical constraints
- Accuracy

# Providing Output Examples

```
{
  "description": "...",
  "name": "...",
  "weapon": "..."
}
```

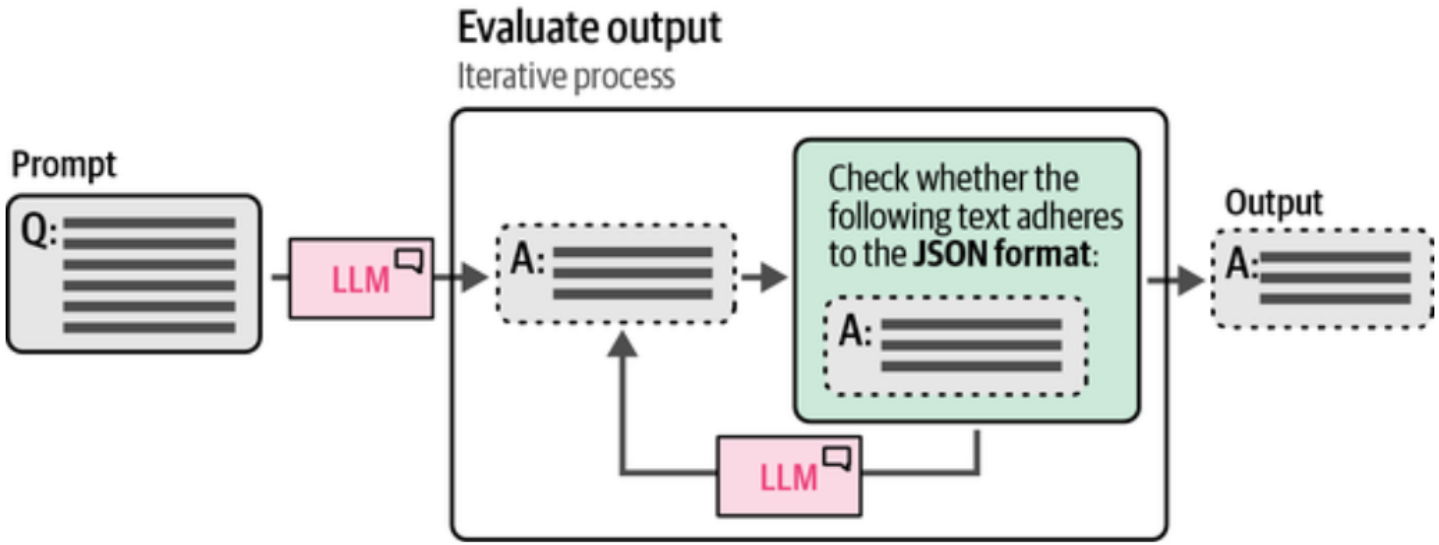
# Output Examples

```python
# One-shot learning: Providing an example of the output structure
one_shot_template = """Create a short character profile for an
RPG game. Make sure to only use this format:

{
  "description": "A SHORT DESCRIPTION",
  "name": "THE CHARACTER'S NAME",
  "armor": "ONE PIECE OF ARMOR",
  "weapon": "ONE OR MORE WEAPONS"
}
"""
one_shot_prompt = [
    {"role": "user", "content": one_shot_template}
]

# Generate the output
outputs = pipe(one_shot_prompt)
print(outputs[0]["generated_text"])
```

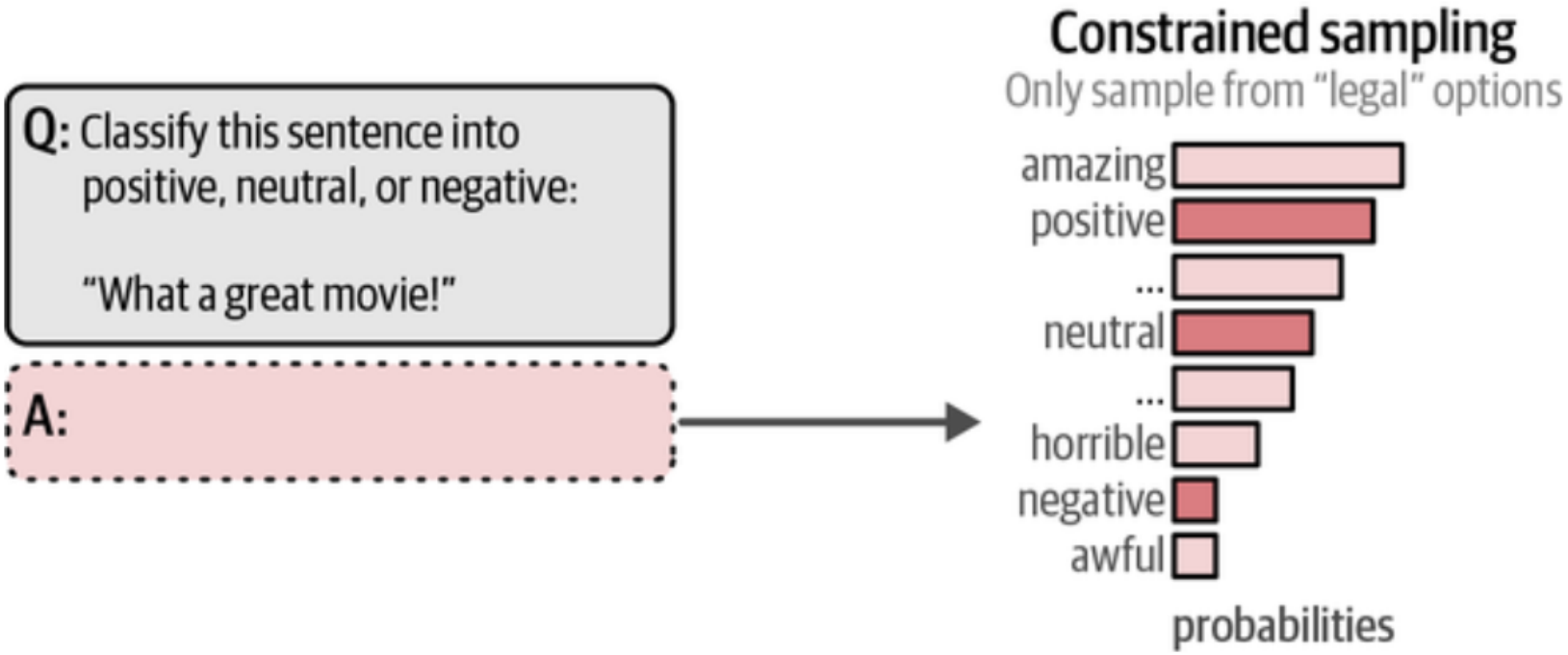
```
 {
   "description": "A cunning rogue with a mysterious past,
 skilled in stealth and deception.",
   "name": "Lysandra Shadowstep",
   "armor": "Leather Cloak of the Night",
   "weapon": "Dagger of Whispers, Throwing Knives"
 }
```

# Grammar Constrained Sampling

- Constrain token selection
- Enforce JSON grammar
- Restrict allowed labels



# Constrained Token Sampling



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